In a blockchain network, miners play a crucial role in securing the network, validating transactions, and adding new blocks to the blockchain. The key tasks of miners in a blockchain are as follows:

- **Transaction Validation:**
  - Miners validate transactions by ensuring that they adhere to the rules of the blockchain network. This includes verifying the digital signatures, confirming that the sender has sufficient funds, and checking the overall consistency of the transaction.
- **Block Creation:**
  - Miners group a set of valid transactions into a new block. They also include a special transaction known as the "coinbase transaction," which rewards the miner with newly created cryptocurrency and transaction fees.
- **Proof-of-Work (PoW) or Consensus Mechanism:**
  - Miners compete to solve a complex mathematical puzzle as part of the consensus mechanism (e.g., Proof-of-Work). The first miner to solve the puzzle broadcasts the solution to the network, providing proof that they expended computational effort. This process is resource-intensive and helps secure the network against malicious activities.
- **Block Propagation:**
  - Once a miner successfully mines a block, they broadcast it to the network. Other nodes in the network validate the block, ensuring that the transactions within it are legitimate and that the miner correctly solved the cryptographic puzzle.
- **Chain Extension:**
  - Miners contribute to extending the blockchain by appending a new block to the existing chain. Each block contains a reference (hash) to the previous block, forming a secure and tamper-resistant chain of blocks.
- **Consensus Maintenance:**
  - Miners play a role in maintaining the decentralized consensus of the network. By participating in the consensus mechanism, they contribute to the agreement on the state of the blockchain, preventing double-spending and ensuring the integrity of the ledger.
- **Transaction Fee Collection:**
  - Miners are typically rewarded with transaction fees paid by users for including their transactions in a block. This serves as an incentive for miners to prioritize transactions with higher fees.
- **Network Security:**
  - Through the Proof-of-Work process, miners contribute to the security of the blockchain network by making it computationally expensive for malicious actors to manipulate the transaction history or disrupt the consensus.